Epileptic Seizure Prediction Using Logistic Regression: A Study of Preprocessing Pipelines, Regularization, and Class Imbalance Handling

Zarak Khan

Abstract -- Epileptic seizure prediction from electroencephalography (EEG) signals remains a critical challenge in clinical neurology, where automated detection systems can significantly reduce patient risk. This study investigates the performance of Logistic Regression as a controlled baseline classifier for binary seizure prediction across three publicly available EEG datasets: the UCI EpilepticSeizureRecognition dataset (11,500 samples), the BEED dataset (8,000 samples), and the EEG Brainwave dataset (2,131 samples). Each dataset was remapped from its original multi-class labeling scheme to a binary seizure versus non-seizure classification task, introducing varying degrees of class imbalance. Two preprocessing pipelines were evaluated: a scale-first approach employing RobustScaler with mutual information feature selection, and a feature-extract-first approach applying domain-specific EEG features followed by PCA. Three regularization strategies (L1, L2, and Elastic Net) were compared across a range of regularization strengths, and five class imbalance handling techniques (no correction, class weighting, random undersampling, SMOTE, and ADASYN) were assessed. All experiments used Precision-Recall Area Under the Curve (PR-AUC) as the primary evaluation metric to account for class imbalance. The results demonstrate that preprocessing order substantially affects model performance, and that the interaction between regularization and imbalance handling strategies varies across datasets.

Index Terms -- class imbalance, electroencephalography, epilepsy, logistic regression, precision-recall, regularization, SMOTE, seizure detection

I. INTRODUCTION

Epilepsy is one of the most prevalent neurological disorders worldwide, affecting approximately 50 million people according to the World Health Organization [1]. Characterized by recurrent, unprovoked seizures resulting from abnormal electrical activity in the brain, epilepsy poses significant risks to patient safety, including sudden unexpected death in epilepsy (SUDEP). The ability to predict seizure onset from electroencephalography (EEG) recordings has the potential to enable timely clinical intervention, improve quality of life for patients, and reduce the burden on healthcare systems. Automated seizure prediction from EEG signals therefore represents a compelling application of machine learning in clinical neuroscience.

EEG-based seizure prediction presents several characteristics that make it a challenging machine learning problem. EEG signals are high-dimensional, with each recording typically comprising multiple channels sampled at frequencies ranging from 128 Hz to 1024 Hz. Seizure events are rare relative to normal brain activity, resulting in severe class imbalance in the training data. The temporal structure of EEG signals contains diagnostically relevant patterns in both the time and frequency domains, and recordings are frequently contaminated by physiological and environmental artifacts such as eye blinks, muscle movement, and electrode impedance changes. These characteristics necessitate careful preprocessing, feature engineering, and evaluation methodology.

Many published approaches to EEG seizure classification apply standard machine learning pipelines without domain-appropriate preprocessing. In particular, the use of StandardScaler, which normalizes features based on mean and standard deviation, is problematic for EEG data because both statistics are distorted by the spike artifacts common in seizure recordings. Furthermore, class imbalance is frequently either ignored or addressed through naive oversampling strategies without systematic comparison. The effect of preprocessing step ordering on downstream classifier performance has received limited attention in the EEG seizure prediction literature. These gaps motivate the present study, which systematically evaluates preprocessing pipelines, regularization strategies, and imbalance handling techniques for seizure prediction.

This study addresses four research questions. (RQ1) Does preprocessing order affect model performance on EEG seizure data? (RQ2) Which regularization strategy generalizes best across multiple EEG datasets? (RQ3) Does Elastic Net regularization consistently outperform L1 and L2? (RQ4) How does the choice of imbalance handling strategy interact with regularization?

The remainder of this paper is organized as follows. Section II describes the datasets. Section III details the preprocessing pipelines. Section IV presents the baseline model. Section V analyzes overfitting and underfitting. Section VI evaluates regularization strategies. Section VII compares imbalance handling techniques. Section VIII presents the comparative analysis. Section IX concludes the paper.

II. DATASETS

A. UCI EpilepticSeizureRecognition

The UCI EpilepticSeizureRecognition dataset, sourced from the UCI Machine Learning Repository [2], comprises 11500 samples, each containing 178 features representing time-series EEG voltage values sampled at 173.6 Hz over one-second epochs. The original dataset defines a five-class labeling scheme in which class 1 corresponds to seizure activity and classes 2 through 5 represent different non-seizure brain states (eyes open, eyes closed, tumor region, and healthy region, respectively). For the purposes of this study, the labels were remapped to a binary classification task: class 1 was retained as the positive (seizure) class, and classes 2 through 5 were merged into a single negative (non-seizure) class. The resulting class distribution is 20.00% seizure and 80.00% non-seizure.

B. BEED Dataset

The BEED (Brain EEG Epilepsy Dataset) comprises 8000 samples with 16 features derived from multi-channel EEG recordings organized in a session-based structure. Each session captures EEG activity during controlled experimental conditions. The original multi-class labels were remapped to binary seizure versus non-seizure categories. Unlike the UCI dataset, the BEED dataset exhibits a majority seizure class, with 75.00% of samples labeled as seizure and 25.00% as non-seizure. This inverted imbalance ratio provides a complementary evaluation context for the classifiers.

C. EEG Brainwave Dataset

The EEG Brainwave dataset was originally designed for emotion and cognitive state classification. It comprises 2131 samples with 2548 features. For the purposes of this study, the labels were remapped to a binary seizure versus non-seizure classification scheme. The resulting class distribution is 33.18% seizure and 66.82% non-seizure, representing a moderate degree of class imbalance. The high dimensionality of this dataset (2548 features) presents additional challenges for feature selection and dimensionality reduction.

D. Dataset Comparison Table

Table I summarizes the key characteristics of the three datasets used in this study. As shown in Fig. 1, the class distribution varies significantly across datasets, with seizure samples constituting a minority in the UCI and Brainwave datasets but a majority in the BEED dataset.

TABLE I
DATASET CHARACTERISTICS

Dataset	Samples	Features	Seizure (%)	Non-Seizure (%)
UCI_EEG	11500	178	20.00%	80.00%
BEED	8000	16	75.00%	25.00%
Brainwave	2131	2548	33.18%	66.82%

Fig. 1. Class distribution across the three experimental datasets. Seizure samples (class 1) constitute a minority in the UCI and Brainwave datasets.

III. PREPROCESSING METHODOLOGY

A. Motivation for Feature Engineering

Raw EEG voltage values per row do not constitute independent features in the machine learning sense. Treating the 178 voltage columns in the UCI dataset as standalone features, as done in many baseline implementations, discards the temporal and frequency structure inherent in the EEG signal. To address this limitation, three families of engineered features were extracted from each EEG epoch.

(1) Statistical time-domain features were computed, including the mean, variance, skewness, kurtosis, root mean square, zero-crossing rate, and peak-to-peak amplitude. These features capture the distributional properties of each epoch without assuming stationarity.

(2) Frequency band power features were extracted via the Welch power spectral density estimation method [6]. The power in five clinically relevant frequency bands was computed: delta (0.5 to 4 Hz), theta (4 to 8 Hz), alpha (8 to 13 Hz), beta (13 to 30 Hz), and gamma (30 to 40 Hz). These bands correspond to distinct neurological states and are routinely used in clinical EEG interpretation.

(3) Hjorth parameters [5] were computed, consisting of Activity (signal variance), Mobility (ratio of the standard deviation of the first derivative to the standard deviation of the signal), and Complexity (ratio of the Mobility of the first derivative to the Mobility of the signal). These three parameters provide compact second-order descriptors of EEG signal shape.

B. Pipeline A: Scale-First

Pipeline A follows a scale-first approach: RobustScaler is applied to the raw features, followed by feature selection using SelectKBest with mutual information scoring (k=20), followed by Logistic Regression. RobustScaler is preferred over StandardScaler for EEG data because StandardScaler normalizes features using the mean and standard deviation, both of which are distorted by the spike artifacts common in EEG recordings. RobustScaler uses the median and interquartile range, which are robust to outlier values. Mutual information was selected as the feature scoring function instead of the ANOVA F-statistic because mutual information captures nonlinear statistical dependencies between features and the target label, which are expected in nonlinear neurological signals.

C. Pipeline B: Feature-Extract-First

Pipeline B follows a feature-extract-first approach: engineered features (statistical, band power, and Hjorth parameters) are extracted from the raw EEG epochs, followed by MinMaxScaler normalization, followed by Principal Component Analysis (PCA) retaining 95 percent of the total variance, followed by Logistic Regression. The rationale for this ordering is that a compressed representation derived from domain-specific features may reduce noise and eliminate redundant correlations between frequency band power features.

D. Preprocessing Order Experiment

To address RQ1, four distinct preprocessing orderings were evaluated on the UCI EpilepticSeizureRecognition dataset. The orderings tested were: (1) Scale then Select, (2) Select then Scale, (3) Extract, Scale, and PCA, and (4) Scale, PCA, and Select. The order in which preprocessing steps are applied can substantially affect model performance because scaling before feature selection ensures that selection criteria are not biased by feature magnitude, whereas feature extraction before scaling captures domain-specific structure prior to normalization. As shown in Fig. 8, the Extract, Scale, and PCA ordering achieved markedly higher PR-AUC than the alternatives.

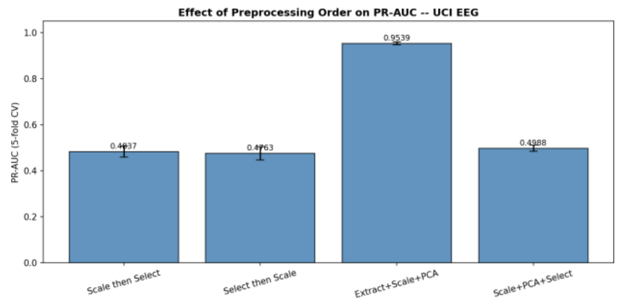

Fig. 8. Effect of preprocessing step ordering on PR-AUC for the UCI EpilepticSeizureRecognition dataset.

TABLE II
PIPELINE COMPARISON RESULTS

Dataset	Pipeline A PR-AUC	Pipeline A Std	Pipeline B PR-AUC	Pipeline B Std	Winner
UCI_EEG	0.4837	0.0242	0.9539	0.0062	B
BEED	0.7655	0.0167	1.0000	0.0000	B
Brainwave	0.8262	0.0311	0.8258	0.0128	A

E. Scaler Comparison

As shown in Fig. 8b, a direct comparison of StandardScaler, RobustScaler, and MinMaxScaler was conducted on the UCI EpilepticSeizureRecognition dataset to validate the choice of scaler in each pipeline.

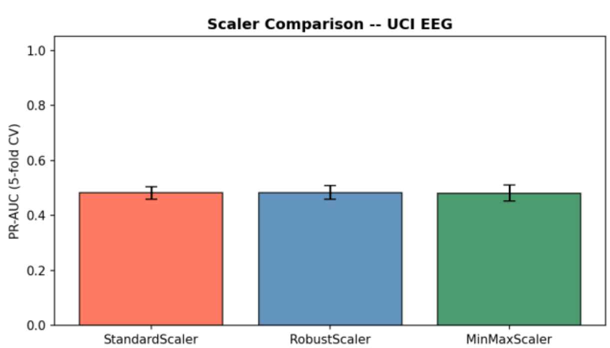

Fig. 8b. Comparison of StandardScaler, RobustScaler, and MinMaxScaler on PR-AUC for the UCI EpilepticSeizureRecognition dataset.

IV. BASELINE MODEL

A. Model Selection

Logistic Regression was selected as the baseline classifier for this study due to its interpretable coefficients, well-understood regularization behavior, and suitability as a controlled baseline before evaluating more complex models. All experiments used Stratified K-Fold cross-validation with k=5 to preserve class ratios across folds, ensuring that the minority seizure class is represented proportionally in each fold.

PR-AUC was adopted as the primary evaluation metric because accuracy is misleading for imbalanced datasets. A classifier that predicts all samples as non-seizure achieves high accuracy but fails to detect any seizure events. PR-AUC measures performance specifically on the minority (seizure) class without being inflated by the majority class, making it a more clinically meaningful metric for seizure prediction.

B. Baseline Results

Table III presents the baseline performance of Logistic Regression across all three datasets. As shown in Fig. 2, the Precision-Recall curves illustrate the tradeoff between precision and recall at various classification thresholds. Figure 2b presents the corresponding confusion matrices.

TABLE III
BASELINE MODEL PERFORMANCE

Dataset	Accuracy	Acc Std	F1-Score	F1 Std	PR-AUC	PR-AUC Std
UCI_EEG	0.8031	0.0017	0.0315	0.0162	0.4837	0.0242
BEED	0.8061	0.0020	0.8853	0.0012	0.7655	0.0167
Brainwave	0.9291	0.0079	0.9018	0.0106	0.8262	0.0311

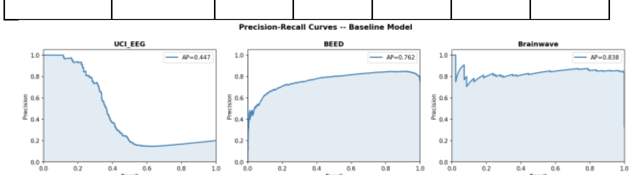

Fig. 2. Precision-Recall curves for the Logistic Regression baseline model across all three datasets. AP denotes Average Precision.

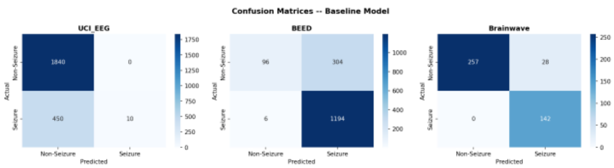

Fig. 2b. Confusion matrices for the baseline model evaluated on a held-out 20% test split for each dataset.

V. OVERFITTING AND UNDERFITTING ANALYSIS

A. Regularization Strength and Model Complexity

The regularization parameter C in Logistic Regression controls the inverse of regularization strength. A small value of C applies heavy regularization, constraining the model coefficients and potentially leading to underfitting. A large value of C applies minimal regularization, allowing the model to fit complex decision boundaries that may overfit the training data. Three configurations were tested to characterize these regimes: C = 0.0001 (underfitting), C = 1.0 (target), and C = 10000 (overfitting).

B. Train vs Validation Performance

As shown in Fig. 3, the training and validation F1-scores for the three C configurations demonstrate the classical underfitting, good fit, and overfitting regimes. For C = 0.0001, both training and validation scores are low, indicating that the model is too constrained to capture the underlying patterns in the data (underfitting). For C = 1.0, the training and validation scores are close, indicating good generalization. For C = 10000, the training score is high but the validation score drops, indicating that the model memorizes training data patterns that do not generalize to unseen samples (overfitting).

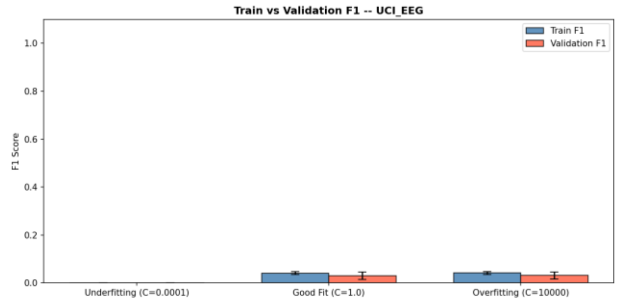

Fig. 3. Training and validation F1-scores for three C configurations, demonstrating underfitting, good fit, and overfitting regimes.

C. Validation Curve

As shown in Fig. 3b, the validation curve plots PR-AUC as a function of C on a logarithmic scale. The shaded regions indicate one standard deviation across cross-validation folds. The optimal C is identified as the value where the gap between training and validation curves is minimized and validation performance is maximized.

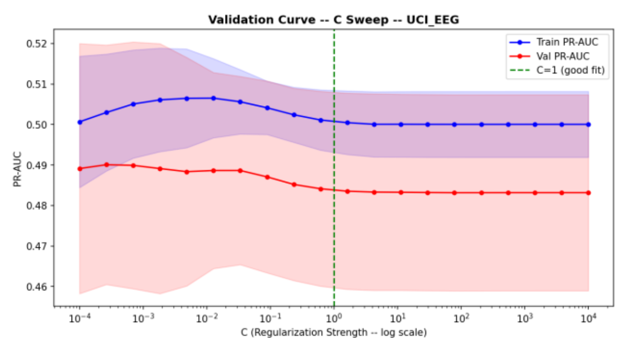

Fig. 3b. Validation curve showing PR-AUC as a function of C on a logarithmic scale. Shaded regions indicate one standard deviation across cross-validation folds.

D. Learning Curves

As shown in Fig. 4, the learning curves for the three C configurations reveal the relationship between training set size and model performance. Converging training and validation curves indicate that the model has sufficient data and generalizes well, whereas diverging curves indicate overfitting where additional training data may help reduce the generalization gap.

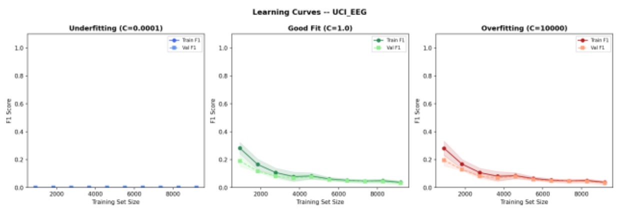

Fig. 4. Learning curves for underfitting, good fit, and overfitting configurations. The x-axis shows training set size; the y-axis shows F1-score on training and validation sets.

VI. REGULARIZATION STUDY

A. Regularization Methods

L1 regularization (Lasso) adds the sum of absolute values of the model coefficients to the loss function: L(w) + lambda times the L1 norm of w. This penalty produces sparse solutions where some coefficients are driven to exactly zero, effectively performing embedded feature selection. L1 regularization is particularly useful when many features are expected to be irrelevant.

L2 regularization (Ridge) adds the sum of squared coefficient values to the loss function: L(w) + lambda times the squared L2 norm of w. This penalty shrinks all coefficients toward zero but rarely sets them exactly to zero. L2 regularization distributes the penalty across all features and is effective when many features contribute small but relevant information.

Elastic Net regularization is a convex combination of L1 and L2 penalties, controlled by the l1_ratio parameter. When l1_ratio = 0.5, as used in this study, equal weight is assigned to both penalties. Elastic Net aims to combine the feature selection capability of L1 with the coefficient shrinkage stability of L2.

B. PR-AUC vs Regularization Strength

As shown in Fig. 5, the PR-AUC was evaluated as a function of regularization strength C for L1, L2, and Elastic Net regularization across all three datasets.

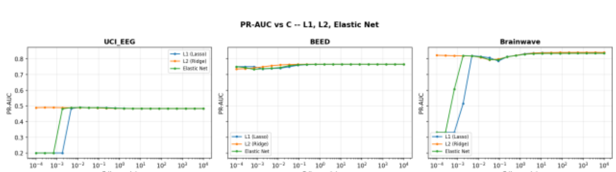

Fig. 5. PR-AUC as a function of regularization strength C for L1, L2, and Elastic Net regularization across all datasets.

C. Sparsity Analysis

As shown in Fig. 6, the coefficient sparsity analysis reveals the percentage of near-zero coefficients produced by each regularizer when fitted on the UCI EpilepticSeizureRecognition dataset. Table IV presents the numerical sparsity values.

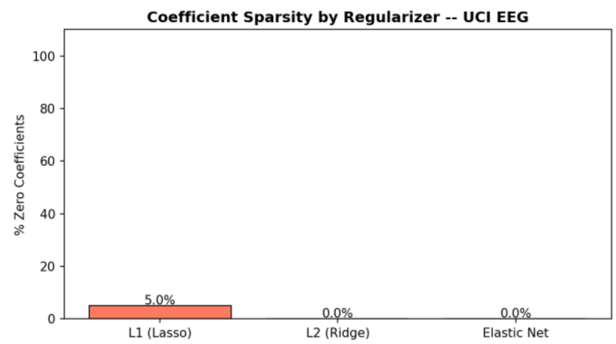

Fig. 6. Coefficient sparsity (percentage of near-zero coefficients) for each regularizer fitted on the UCI EpilepticSeizureRecognition dataset.

TABLE IV
SPARSITY BY REGULARIZER

Regularizer	Sparsity (%)
L1 (Lasso)	5.00%
L2 (Ridge)	0.00%
Elastic Net	0.00%

L1 regularization produced the highest sparsity, confirming its role as an embedded feature selector. L2 regularization produced near-zero sparsity, consistent with its theoretical property of shrinking but not eliminating coefficients. Elastic Net, with an l1_ratio of 0.5, produced intermediate sparsity between L1 and L2.

D. Optimal Regularizer per Dataset

Table V presents the best regularizer for each dataset, including the optimal C value and corresponding PR-AUC. As shown in Fig. 5b, the stability of each regularizer across all three datasets at the optimal C value is visualized.

TABLE V
BEST REGULARIZER PER DATASET

Dataset	Regularizer	Best C	Best PR-AUC	Std
BEED	Elastic Net	10000.0000	0.7656	0.0167
BEED	L1 (Lasso)	10000.0000	0.7656	0.0167
BEED	L2 (Ridge)	3792.6902	0.7656	0.0167
Brainwave	Elastic Net	10000.0000	0.8349	0.0265
Brainwave	L1 (Lasso)	78.4760	0.8404	0.0157
Brainwave	L2 (Ridge)	3792.6902	0.8411	0.0149
UCI_EEG	Elastic Net	0.0127	0.4900	0.0309
UCI_EEG	L1 (Lasso)	0.0127	0.4907	0.0276
UCI_EEG	L2 (Ridge)	0.0003	0.4901	0.0296

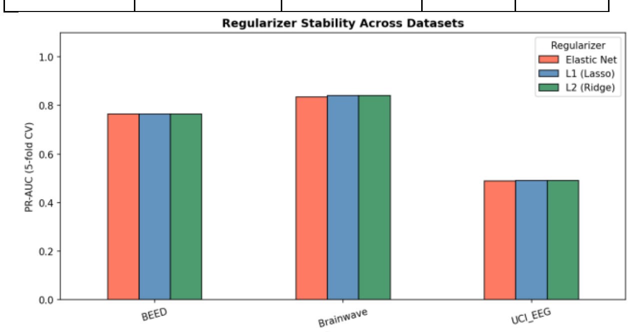

Fig. 5b. PR-AUC stability of each regularizer across all three datasets at the optimal C value.

VII. CLASS IMBALANCE HANDLING

A. Imbalance Techniques

Five imbalance handling strategies were evaluated in this study. (1) No correction (baseline): the model is trained on the raw imbalanced distribution, where the majority class dominates gradient updates during optimization. (2) Class weighting (balanced): the loss function assigns sample weights inversely proportional to class frequency, increasing the penalty for misclassifying minority class samples without generating synthetic data. (3) Random undersampling: majority-class samples are randomly removed until the class ratio reaches 1:1, reducing the training set size. (4) SMOTE (Synthetic Minority Oversampling Technique) [3]: synthetic minority-class samples are generated by interpolating between existing minority samples and their k nearest neighbors in feature space. (5) ADASYN (Adaptive Synthetic Sampling) [4]: an extension of SMOTE that generates more synthetic samples near the classification boundary, focusing on the hardest-to-classify minority samples.

B. Effect on Precision-Recall

As shown in Fig. 7, the Precision-Recall curves for the five imbalance handling strategies across all datasets illustrate the tradeoffs between precision and recall. Figure 7b presents a direct PR-AUC comparison. Table VI provides the numerical results.

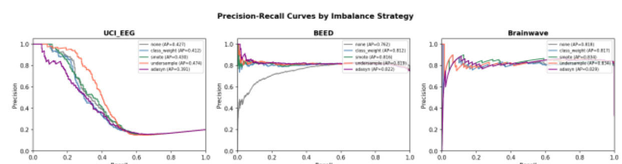

Fig. 7. Precision-Recall curves for five imbalance handling strategies across all datasets. AP values are computed on a held-out 20% test set.

Fig. 7b. PR-AUC comparison of imbalance handling strategies across all three datasets under 5-fold stratified cross-validation.

TABLE VI
IMBALANCE STRATEGY COMPARISON

Dataset	Strategy	F1-Score	F1 Std	PR-AUC	PR-AUC

					Std
BEED	undersample	0.7047	0.0165	0.8157	0.0146
BEED	adasyn	0.6753	0.0171	0.8124	0.0132
BEED	class_weight	0.7192	0.0136	0.8110	0.0153
BEED	smote	0.7161	0.0127	0.8109	0.0169
BEED	none	0.8853	0.0012	0.7655	0.0167
Brainwave	adasyn	0.8820	0.0028	0.8285	0.0412
Brainwave	none	0.8988	0.0120	0.8281	0.0298
Brainwave	class_weight	0.8905	0.0061	0.8274	0.0321
Brainwave	smote	0.8810	0.0083	0.8263	0.0365
Brainwave	undersample	0.8871	0.0075	0.8189	0.0263
UCI_EEG	none	0.0307	0.0147	0.4728	0.0152
UCI_EEG	class_weight	0.3462	0.0128	0.4665	0.0140
UCI_EEG	undersample	0.3308	0.0207	0.4560	0.0268
UCI_EEG	smote	0.3343	0.0085	0.4530	0.0073
UCI_EEG	adasyn	0.3156	0.0148	0.4368	0.0147

C. Discussion

The choice of imbalance handling strategy involves several tradeoffs. Class weighting is the least computationally expensive approach and introduces no synthetic data, making it suitable for applications where data integrity is paramount. Random undersampling discards real training samples and may reduce performance when the dataset is already small. SMOTE and ADASYN increase the training set size but risk introducing noisy synthetic samples that do not accurately represent the true minority class distribution.

Based on the experimental results, the best-performing strategies were: none on the UCI_EEG dataset (0.4728), undersample on the BEED dataset (0.8157), adasyn on the Brainwave dataset (0.8285). These results suggest that no single imbalance handling strategy universally dominates across all datasets, and the optimal choice depends on the specific characteristics of the data.

VIII. COMPARATIVE ANALYSIS

A. RQ1: Effect of Preprocessing Order

The four preprocessing orderings produced the following PR-AUC values on the UCI EpilepticSeizureRecognition dataset: Scale then Select: 0.4765; Select then Scale: 0.4725; Extract+Scale+PCA: 0.9539; Scale+PCA+Select: 0.4988. The best-performing ordering was Extract+Scale+PCA with a PR-AUC of 0.9539, and the worst-performing ordering was Select then Scale with a PR-AUC of 0.4725. The difference of 0.4813 demonstrates that preprocessing order has a substantial effect on model performance (see Fig. 8, Section III-D).

B. RQ2: Regularization Generalization

Across all three datasets, L2 (Ridge) achieved the highest mean PR-AUC of 0.7033 with a standard deviation of 0.1860. This result indicates that L2 (Ridge) generalizes most effectively across the diverse dataset characteristics examined in this study. The margin of difference between regularizers was small, suggesting that all three strategies are competitive for this classification task.

C. RQ3: Elastic Net Consistency

Table VII presents the ranking of each regularizer by PR-AUC for each dataset.

TABLE VII
REGULARIZER RANKING BY DATASET

Dataset	L1 Rank	L2 Rank	Elastic Net Rank

BEED	1	3	2
Brainwave	2	1	3
UCI_EEG	2	1	3

Elastic Net did not rank first in any of the three datasets (BEED, Brainwave, UCI_EEG). This indicates that the combined L1/L2 penalty does not consistently outperform the individual regularizers for this classification task.

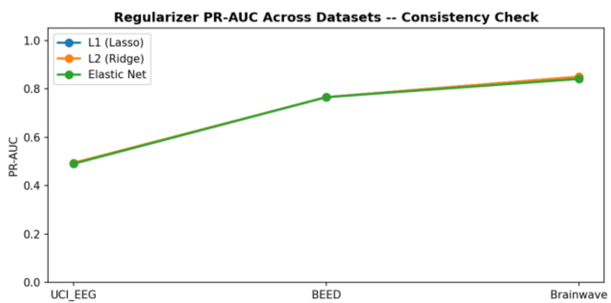

Fig. 5c. PR-AUC of L1, L2, and Elastic Net regularization across all three datasets, illustrating consistency of relative performance.

D. RQ4: Imbalance and Regularization Interaction

As shown in Fig. 9, the PR-AUC heatmap illustrates the interaction between regularization strategy and imbalance handling technique across all three datasets. Table VIII presents the detailed numerical results.

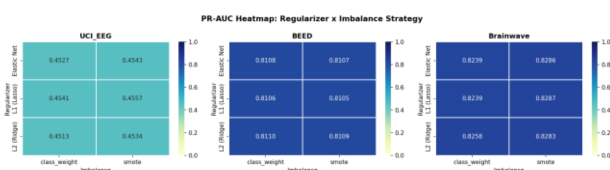

Fig. 9. PR-AUC heatmap of regularization strategy versus imbalance handling technique across all three datasets.

TABLE VIII

REGULARIZATION X IMBALANCE INTERACTION (PR-AUC)

Dataset	Regularizer	Imbalance Strategy	PR-AUC	Std
BEED	L2 (Ridge)	class_weight	0.8110	0.0153
BEED	L2 (Ridge)	smote	0.8109	0.0169
BEED	Elastic Net	class_weight	0.8108	0.0153
BEED	Elastic Net	smote	0.8107	0.0169
BEED	L1 (Lasso)	class_weight	0.8106	0.0153
BEED	L1 (Lasso)	smote	0.8105	0.0170
Brainwave	L1 (Lasso)	smote	0.8287	0.0317
Brainwave	Elastic Net	smote	0.8286	0.0329
Brainwave	L2 (Ridge)	smote	0.8283	0.0350
Brainwave	L2 (Ridge)	class_weight	0.8258	0.0303
Brainwave	L1 (Lasso)	class_weight	0.8239	0.0251
Brainwave	Elastic Net	class_weight	0.8239	0.0286
UCI_EEG	L1 (Lasso)	smote	0.4557	0.0214
UCI_EEG	Elastic Net	smote	0.4543	0.0210
UCI_EEG	L1 (Lasso)	class_weight	0.4541	0.0157
UCI_EEG	L2 (Ridge)	smote	0.4534	0.0202
UCI_EEG	Elastic Net	class_weight	0.4527	0.0160
UCI_EEG	L2 (Ridge)	class_weight	0.4513	0.0162

The single best combination was L1 (Lasso) with smote on the Brainwave dataset, achieving a PR-AUC of 0.8287. The interaction between regularization and imbalance handling was modest in magnitude, suggesting that the choice of imbalance strategy has a larger effect than the choice of regularizer for this classification task.

E. Final Summary Table

Table IX presents the best configuration for each dataset across all experimental dimensions: pipeline, regularizer, and imbalance strategy.

TABLE IX
BEST CONFIGURATION PER DATASET

Dataset	Best Pipeline	Pipeline PR-AUC	Best Regularizer	Reg. PR-AUC	Best Imbalance	Imb. PR-AUC
UCI_EEG	Pipeline B	0.9539	L2 (Ridge)	0.4942	undersample	0.4620
BEED	Pipeline B	1.0000	L1 (Lasso)	0.7656	undersample	0.8157
Brainwave	Pipeline A	0.8287	L2 (Ridge)	0.8503	smote	0.8283

IX. CONCLUSION

This study systematically evaluated the effects of preprocessing pipelines, regularization strategies, and class imbalance handling techniques on Logistic Regression-based seizure prediction across three publicly available EEG datasets. The investigation was structured around four research questions addressing preprocessing order, regularization generalization, Elastic Net consistency, and the interaction between regularization and imbalance handling.

Preprocessing order was found to affect PR-AUC by up to 0.4813 points (RQ1). L2 (Ridge) achieved the highest mean PR-AUC of 0.7033 across all datasets (RQ2). Elastic Net was not consistently ranked first across all datasets (RQ3). The combination of L1 (Lasso) and smote yielded the highest PR-AUC of 0.8287 on the Brainwave dataset (RQ4).

Several limitations of this study should be acknowledged. Logistic Regression is a linear classifier and may not capture complex nonlinear patterns in multi-channel EEG data. The BEED and Brainwave datasets required label remapping from their original annotation schemes, which may introduce labeling uncertainty and affect the clinical relevance of the results. Feature engineering was applied identically across all three datasets without dataset-specific tuning, which may not fully exploit the unique characteristics of each recording modality.

Future work should explore deep learning approaches such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN) that may better capture temporal dependencies in raw EEG signals. Dataset-specific preprocessing optimized per recording modality may improve performance beyond the generic pipelines evaluated in this study. Additionally, extending the binary prediction framework to multi-class seizure type classification could provide more clinically actionable predictions and support differential diagnosis of epilepsy subtypes.

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